Neural net based determination of generator-shedding requirements in electric power systems

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Indexing terms: Electric power systems, Neural nets

Abstract: This paper presents an application of artificial neural networks (ANN) in support of a decision-making process by power system operators directed towards the fast stabilisation of multi-machine systems. The proposed approach considers generator shedding as the most effective discrete supplementary control for improving the dynamic performance of faulted power systems and preventing instabilities. The sensitivity of the transient energy function (TEF) with respect to changes in the amount of dropped generation is used during the training phase of ANNs to assess the critical amount of generator shedding required to prevent the loss of synchronism. The learning capabilities of neural nets are used to establish complex mappings between fault information and the amount of generation to be shed, suggesting it as the control signal to the power system operator.

List of symbols

NG = number of generator nodes
P_{mi} = mechanical power of ith generator
$P_{ei}(0^+)$ = electrical output (active power) of ith generator immediately after occurrence of a fault
E_i = electromotive force behind transient reactance of ith generator
G_{ij}, B_{ij} = components of ijth elements of reduced short circuit admittance matrix for network
M_i $= \dfrac{T_{Ji} S_{ni}}{\omega_s}$ = inertia constant of ith synchronous machine
T_{Ji} = inertia time constant
S_{ni} = rated apparent power
ω_s = synchronous speed of rotation
M_{COI} $= \sum_{i=1}^{NG} M_i$
δ_i = rotor angle of ith generator relative to synchronously rotating reference frame
ω_i = angular velocity of rotor of ith generator relative to synchronous velocity
δ_0 $= \dfrac{1}{M_T} \sum_{i=1}^{NG} M_i \delta_i \quad \theta_i = \delta_i - \delta_0 \quad \tilde{\omega}_i = \omega_i - \omega_0$
C_{ij} $= E_i E_j B_{ij}$; $D_{ij} = E_i E_j G_{ij}$
θ^s = postdisturbance stable equilibrium point

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1 Introduction

The transient stability control and the dynamic stability control aim to improve the steady-state stability and transient stability in a co-ordinated manner [1, 2]. The role of transient stability control is to dampen power system swings after severe disturbances and imbalances of the mechanical and electrical power have occurred, while the dynamic stability control takes over small system oscillations due to a temporary imbalance between the mechanical and electrical torques of the power plant.

Generator shedding is one of the most effective discrete supplementary control means for emergency control especially in predominantly hydroelectric power systems with a relatively large installed capacity and long transmission lines to the main consuming areas In such systems, a generator-shedding strategy is usually combined with other discrete supplementary controls (high speed static excitation, braking resistor and series compensation). An analytical method for estimating the amount of generation-shedding required to prevent the loss of synchronism has been proposed [3].

The block emergency prevention control (BSPC) applied in Japan [4] performs an optimisation routine to decide whether generator tripping, loop separation or system splitting should be chosen to prevent the loss of synchronism. Generator tripping has also been proposed as an emergency control measure [5–7] based on the energy function method.

2 Artificial neural networks

Artificial neural networks (ANN) [8, 13], also known as parallel distributed processing systems or connectionist

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networks, are based on the methods of information processing which are understood to exist in the brain. They consist of a large number of simple processing units, massively interconnected. Such processing architectures have the capability to create their own subsymbolic representations, to learn, to memorise, and to recall associatively.

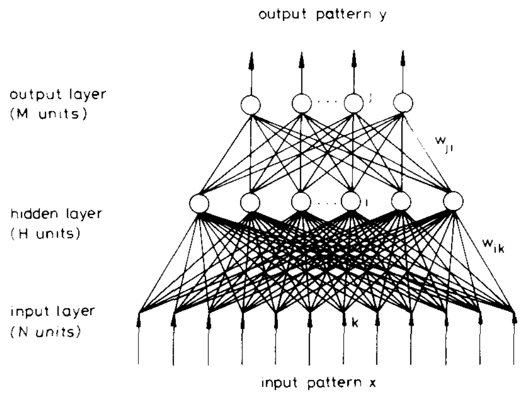


Fig. 1 *Feedforward neural network*

A typical feedforward neural network with a single hidden layer is illustrated in Fig. 1. It has been shown that such networks can approximate any nonlinear function to an arbitrary accuracy [9–11]. The idea underlying the design of these networks is that the information provided to the network input is being recorded in the form of an internal representation which, in turn, drives the network outputs either by itself or in combination with the original input features.

Rumelhart *et al.* [8] have demonstrated that a Perceptron-type feedforward layered machine with one internal layer could indeed train itself autonomously as desired, if analytic functions were used for activation at the network nodes, and if a generalised delta rule (GDR) was used to change the interconnecting weights, activation functions and thresholds until the proper recognition capability had been attained. Supervised learning may be treated as a mechanism which, when presented with a sequence of class labelled patterns, learns an internal structure which allows it to generalise and classify other patterns correctly. The input signals come either from the environment or from the outputs of other processing units and from the input pattern vector. A typical neuron, which is the elementary processor unit of a neural net, utilises the logistic activation function as shown in Fig. 2. The parameter θ_j is called a threshold and determines the transition region of the function, while the parameter θ_0 determines the abruptness of the transition. The weight corresponding to each unit, a collection of which forms the weight vector $\boldsymbol{w}$, where w_i represents the connection strength for the ith input is also shown. The activation or total input for a unit in layer j is defined as $\mathrm{net}_j = \sum_{i=1}^{n} w_{ji} o_i$.

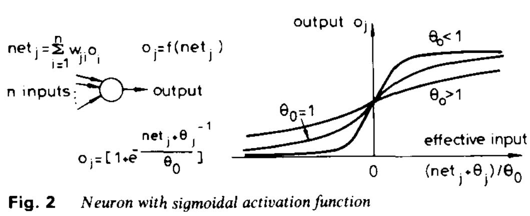


Fig. 2 *Neuron with sigmoidal activation function*

The activation function determines the output value of a processing unit. The sigmoidal activation function, that is a hybrid of the ramp and the step activation functions, shown in Fig. 2, provides a graded nonlinear response to the input signal. It has the form:

$$f \triangleq \frac{1}{1 + e^{-(\mathrm{net}_j + \theta_j)/\theta_0}} \tag{1}$$

The sigmoidal function has the important property of exponential maps in that its first derivative can be expressed in terms of the function itself. This function yields an output which varies continuously from 0 to 1. The quantity θ_j serves as a 'threshold' and positions the transition region of the f function.

Feedforward networks operate in two distinctive phases, the learning phase and the consulting phase. In the learning phase the patterns are fed into the input layer and propagated forward to the output layer, where the actual outputs are compared with the desired ones and the error is obtained. The error is propagated backwards through the network adjusting the weights of each layer. Adjustments of weights are made by first updating the hidden-to-output weights, and computing the error signals at the output layer. These terms are then propagated back to the hidden layer to compute the error signals which are used to update the input-to-hidden weights.

The functional link net (FLN) introduced by Pao [13] represents the new network architecture that allows unsupervised learning, supervised learning and associative retrieval to be carried out with the same net configuration and with the same data structure. The supervised learning paradigm, can simplify the net architecture and also significantly reduce the time to convergence. The basic idea behind a functional link net is the use of links for effecting information processing transformations. The concept of a functional link is illustrated in Fig. 3.

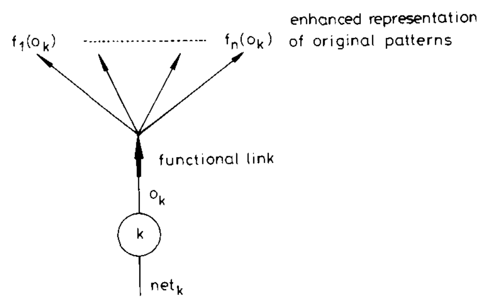


Fig. 3 *Schematic illustration of a functional link*

o_k output of node k
net_k net input to node k

The essential action is therefore the generation of an enhanced pattern to be used in place of the actual pattern. The idea is that the activation of node k offers the possibility that different additional processes $f_1(o_k)$, $\ldots, f_n(o_k)$ may also be activated through a functional link. The generalised delta rule net operates on the assumption that the nonlinear vector-to-vector mapping can be represented by a sequence of linear matrix multiplications, each of which is followed by a nonlinear activation function transformation. The form of the activation function is fixed to be one of several standard varieties (e.g., a sigmoid) and only the values of the threshold parameter are varied in the learning process. In contrast to this, in the FLN approach we focus on learning the functional form of approximating mapping the unknown nonlinear functions o_k in terms of an expansion with unknown coefficients which it determines through delta rule learning. In this way the generation of an enhanced pattern to be used in place of an actual pattern is performed. There is a mathematical basis also pragmatic evidence that supervised learning can be successfully achieved using a flat net and the delta rule if the enhancements are carried out correctly. This is in contrast to the conventional use of a net with the obligatory hidden layers and the generalised delta rule.

There are two models for the functional link net, the functional expansion model and the tensor (or outer product) model. It should be noted that these models can be used in tandem to good effect. In the functional expansion model, the FLN acts on each node singly (typically through orthonormal basis functions such as [sin $k\pi x$, cos $k\pi x$; $k = 1, 2, \ldots$]). The net effect is to map the input pattern onto a higher-dimension pattern space. In the tensor or outer-product model each component of the input pattern multiplies the entire pattern vector such as

$$\{x_i\} \Rightarrow \{x_i, \underset{j \geq i}{x_i x_j}\} \Rightarrow \{x_i, \underset{j \geq i}{x_i x_j}, \underset{k \geq j \geq i}{x_i x_j x_k}\} \Rightarrow \cdots$$

In this case the functional link generates an entire vector from each of the individual components. Augmenting the vector allows the original pattern to be regenerated along with the higher-order terms. It might induce the same set of additional functionalities for each and every node in the input pattern space (see Figs. 4*a* and 4*b*).

The outer-product enhancement procedure is efficient in introducing additional facets of representation which are often sufficient to support the learning process. The degree of enhancement is often low. But the straightforward functional expansion procedure is systematic and is always effective. Experience indicates that a low degree of functional expansion followed by outer-product enhancement is often both efficient and effective.

The mathematical basis for the functional link net has been given by Sobajic [12]. The net configuration is given in Fig. 5.

Let us consider the possibility of learning with a flat net and let there be P patterns each with N elements. For pattern p the pattern components are $x_i^{(p)}$ and the output is $o^{(p)}$. The weighting factors along the links are w_i and the node threshold is θ, these being true for all the patterns $p = 1, 2, \ldots, P$.

Since, in general

$$o = \frac{1}{1 + e^{-[(\text{net} + \theta)/\theta_0]}} \tag{2}$$

where θ_0 is the slope ('temperature') parameter of the sigmoidal curve, θ is the node threshold. Then we have:

$$\text{net} + \theta = \ln\left(\frac{o}{1 - o}\right)^{\theta_0} = z \tag{3}$$

i.e.

$$w_1 x_1^{(p)} + \cdots + w_N x_N^{(p)} + \theta = z^{(p)} \quad p = 1, 2, \ldots, P \tag{4}$$

then for all the P patterns:

$$\begin{bmatrix} x_1^{(1)} & \cdots & x_N^{(1)} & 1 \\ \vdots & & \vdots & 1 \\ \vdots & & \vdots & \vdots \\ \vdots & & \vdots & 1 \\ x_1^{(P)} & \cdots & x_N^{(P)} & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_N \\ \theta \end{bmatrix} = \begin{bmatrix} z^{(1)} \\ \vdots \\ z^{(P)} \end{bmatrix} \tag{5}$$

Thus, finding the weights for a flat net consists of solving a system of simultaneous linear equations for the weights

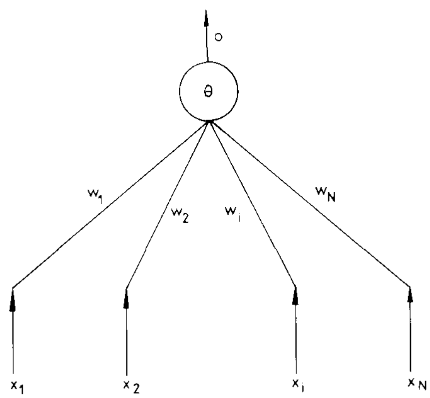
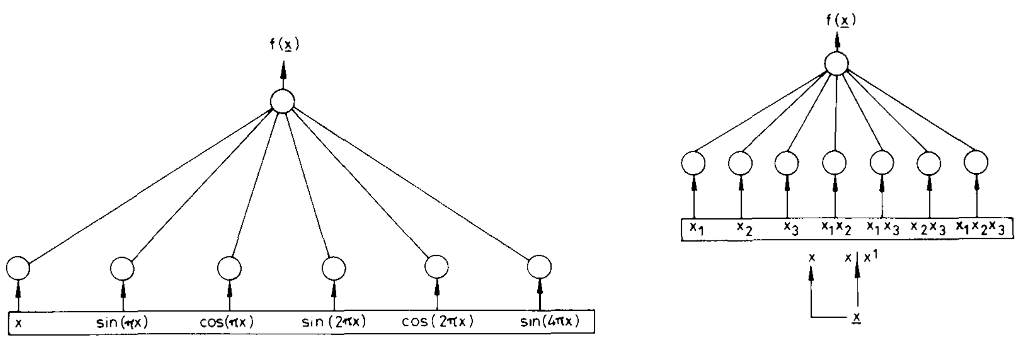


Fig. 5 *Model of functional link net*

w_i and the threshold θ. That is, we need to solve the linear matrix equation

$$\boldsymbol{X}\boldsymbol{w} = \boldsymbol{z} \tag{6}$$



a *b*

Fig. 4 *Functional link net architecture*
a functional expansion model *b* outer product model

where the dimensions of the matrix X are $P \times (N+1)$. We note that if $P = N + 1$ and det $X \neq 0$ then

$$w = X^{-1}z \tag{7}$$

If $P < N + 1$, then we can partition X to obtain a matrix X_F of dimension $P \times P$. We set

$$w_{P+1} = w_{P+2} = \cdots = w_N = 0 \tag{8}$$

and if det $X_F \neq 0$ then

$$w = X_F^{-1}z \tag{9}$$

In the case where $P > N + 1$ the role of the functional link is to augment the number of pattern elements such that we have

$$\begin{array}{c|c} \multicolumn{1}{c}{N+1} & \\ \end{array} {}_{P}\left[X \;\middle|\; X_E \right]^{N_{FL}} \begin{bmatrix} w \\ w_E \end{bmatrix} = [z] \tag{10}$$

where N_{FL} is the dimension of the enhanced patterns X_E is the enhancement portion of the pattern descriptions $X_F = [X|X_E]$ and w_E are the weights for the additional nodes. Because the functional link can generate an infinitely large number of orthonormal functions, N_{FL} can always be made to be equal or greater than P. Thus if $N_{FL} = P$ and det $X_{FL} \neq 0$ then

$$w_{FL} = X_{FL}^{-1}z \tag{11}$$

and an exact flat net solution is obtained. If $N_{FL} > P$, and rank $X_{FL} = P$ we apply eqn. 9. In conclusion, the criterion for the enhancement of the input pattern representation is suggested: a new feature x_{s+1} is used for input pattern enhancement only if

$$\text{rank } X_{s=1} = \text{rank } X_s + 1 \tag{12}$$

where X_s is a $P \times S$ dimensional matrix containing P patterns where each pattern is represented by S features. This analysis indicates that the functional expansion model always yields a flat net solution if a sufficiently large number of additional functions, generated according to expr. (12) is used for the enhancement of the original pattern representation and data compression preprocessing is used to remove redundant patterns.

In contrast to the linear weighting produced by the linear links of the generalised delta rule net, the functional link acts on an element of a pattern or on the entire pattern itself by generating a set of linearly independent functions, then evaluating these functions with the pattern as the argument. In one sense, no new *ad hoc* information has been inserted into the process, however, the representation has been enhanced, and separability becomes possible in the enhanced space. The use of the FLN nets 'flat' architecture with no hidden layers increases the learning rates, simplifies the learning algorithms and allows both supervised and unsupervised learning to be carried out with the same net architecture, with no configuring and no shuffling of data from one net to another.

The investigation of artificial neural networks has covered many topics in interpretation, diagnostics, forecasting, predictive monitoring, control and a variety of other tasks [14–22].

3 Determination of generator-shedding requirements using parallel associative memories

Reliable operation of modern electric power systems requires qualitative improvements in the functionalities of its processing information systems. The understanding of physical phenomena and good communication skills are prerequisites to enable the system operator to create timely and adequate actions in his minute to minute decision-making process. However, the burden of complexity in the process of monitoring, supervision and decision making has increased. To develop efficient processing of large amounts of data gathered through communication channels and with emphasis on avoiding costly human errors, the use of cognitive engineering technologies implemented on parallel distributed computational architecture of neural nets is explored.

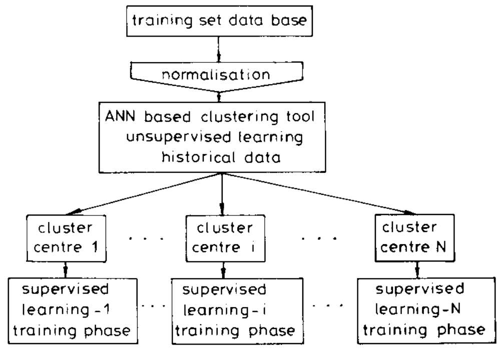


Fig. 6. *Unsupervised/supervised learning concept*

In formulating advanced control strategies we have proposed a system based on neural nets for power system monitoring and diagnosis [14]. The neural net subsystem for the state classification within this concept is structured using unsupervised/supervised learning as shown in Fig. 6. There are several reasons for using the unsupervised learning mode of neural net computing

(*a*) Massive data compression
(*b*) Building hierarchical neural net structures
(*c*) Autonomous feature discovery

The use of unsupervised learning can greatly reduce the dimensionality of the problem and enable subsequent learning paradigms to perform in well defined circumstances. Input patterns are clustered according to similarities discovered among the input features. Taking each cluster, one at a time, we can examine the class label of the patterns to see if they are indeed 'similar' to the extent of their class membership being the same. If they are, then we can synthesise a set of weights which will yield the desired outputs whenever one of those patterns is presented as the input. At the second stage we proceed with a supervised learning paradigm for accurate estimation of output features.

This co-existence between supervised learning and unsupervised learning relieves the net of the burden of trying to associate many dissimilar patterns with the same output. If two groups, which are not alike, happen to end up in the same cluster, the message is that either pattern description is not adequate or the vigilance factor is not stringent enough. Using the unsupervised/supervised concept the supervised learning process is greatly facilitated. It is carried out on the clusterwise data structure rather than on the entire data set.

Because of the extremely fast response and the adaptation capabilities, neural-net based monitoring and diagnosis of the operating conditions is the ideal candidate for the real-time implementation. In real-time operation and control of interconnected systems it is essential to be able to predict the consequences of unforeseen disturbances accurately. In this paper we advance further the ideas of the neural-net based monitoring and diagnosis concept.

We propose an associative memory system for the determination of generator shedding requirements (AMGS) intended to assist the operator's decision process in power system dynamics characterisation. The proposed approach considers generator shedding as the most effective control action for preventing transient instability of power systems exposed to severe faults. A schematic diagram of the AMGS is shown in Fig. 7.

system [21] according to

$$F_i = \left\{ \frac{1}{M_i} [P_{mi} - P_{ei}(0^+)] - \frac{1}{M_{COI}} \left[\sum_{i=1}^{NG} P_{mi} - \sum_{i=1}^{NG} P_{ei}(0^+) \right] \right\} \frac{Y_{if}}{Y_{ii}} \qquad i = 1, \ldots, NG \quad (13)$$

where Y_{if} is the admittance distance from generator 'i' to the faulted bus and Y_{ii} is a self-admittance of the generator bus 'i'.

The proper selection of system description parameters has a profound effect on the efficiency and 'correctness' of

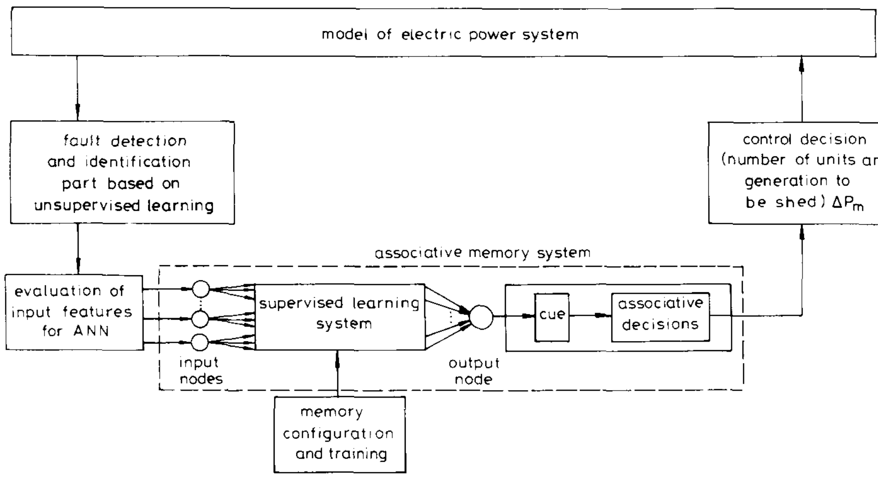


Fig. 7 *Use of associative memory system to determine generator shedding requirements (AMGS)*

The role of the fault detection and identification unit (FDI) is to calculate the electrical output powers P_{fi} of generators ($i = 1, \ldots, NG$) and mechanical inputs P_{mi} during a fault and to identify the type and location of fault using the unsupervised learning concept. Individual accelerations of generators obtained from mechanical and electrical powers are used as features for defining input patterns. The unsupervised learning system will self-organise incoming data based on similarities discovered among the input patterns. The range of label attributes (individual accelerations) in each cluster indicates which associative neural net should be taken in each instance. Neural nets are separately trained to provide specific generator shedding control actions which are most efficient in dampening power system swings after the occurrence of a certain disturbance. The potential use of the unsupervised learning paradigm in screening system disturbances is clear. Based on a rough classification, it can inform and improve decision making in real-time power system operation and planning, providing new, enhanced insights into system conditions. Focusing on each cluster separately, a FLN architecture of supervised learning is designed at the second stage to estimate accurately the required amount of shed generation needed to prevent loss of synchronism. Information about faults, with the information about topological observability and network connectivity is used to calculate input features for a neural net associative memory

an associative memory. One may approach this task in two ways. One may look at a 'large picture' that contains numerous 'raw' measurements and let the neural net system derive or detect the important ones [18]. Or, one can exercise engineering judgment and produce a set of derived features immediately. Each approach has it advantages and disadvantages. 'Looking at a large picture' is recommended if little or no knowledge is available on the process and data compression and feature extraction are strongly influenced by the quality of the data. Pragmatic features yield compact and easily trainable memories. Yet, the danger is that some important parameters may be overlooked. This is normally verified when the memory 'correctness' is evaluated and the feature set is augmented if necessary.

The AMGS system presented in Fig. 7 consists of two modules. The first comprises a supervised learning net, the inputs and outputs (targets) of which are defined in terms of F_i coefficients and shed generation, respectively. As a result of the learning process, a unique set of weights and thresholds is established on the basis of presented data collected under various operating conditions, fault types and transmission network topologies. During a consulting phase, a trained net is able to synthesise appropriate control measures, i.e. amounts of shed generation corresponding to new and previously 'unseen' circumstances. It should be noted that the character of the learned map is 'analogue-to-analogue'. The nature of the problem dictates the quantisation of the estimated control signals so that the shed generation amounts are feasible.

The quantisation is carried out according to the nominal apparent power and the number of generator units in the power stations which are selected for generator shedding. The associate decisions or actions are then related to the quantisation intervals. This is the task of a second module, where the output of a supervised learning system serves as a cue for the associative decision retrieval. If the output node of a supervised network causes an associated control action to be activated, a message such as 'shed M units in power station A with mechanical power of X MVA' may be recalled provided that the output falls within the appropriate quantised interval.

The emphasis in the proposed methodology is on providing a fast and reliable tool for a quick assessment of the consequences of a given fault on power system dynamic security and the effects of previously specified generator shedding schemes in preventing transient instability. In real-time conditions the role of the AMGS is to schedule intelligent emergency control actions. In such circumstances the AMGS will receive the required information from a fast responsive monitoring subsystem, (see Fig. 6).

In an offline application (e.g. operator's training simulator) after fault identification postprocessing (using a supervised learning system), only the recommended control action is presented to the operator through the operator's interface (cue and associative decision). Then, only verified and approved control actions will be stored in the memory, to be initiated only if such circumstances repeat during the actual operation.

A single neural network, irrespective of its size, is the least likely candidate for the AMGS concept, remembering that the actual power system operation is characterised by frequent changes of load levels and occasionally network topology. This problem is approached using a hierarchical neural net system in which the first level monitors the fault location and network topology and activates one of many individual neural nets trained for different, ever changing, load levels. Consequently AMGS is structured as a hierarchical decision system where each neural net is trained to respond to a specific generator shedding option [16].

The architecture of a neural net system in real-time power system operation has to be flexible and robust to provide for both fast storage of incoming data sequences, and to maintain its highly accurate predictive capabilities. Thus we propose a system structure based on the unsupervised/supervised learning concept. It should be noted that the behaviour of neural nets is conditioned through the training process. Consequently, changes or modifications in neural net behaviour are obtained through the continuation of the training process and the presentation of new data obtained from new loading conditions and/or transmission network topologies.

An important aspect of the AMGS is its application to large power systems. Practical power systems tend to operate in large interconnections with everyday changes in the load levels and the configuration of generators, transformers and lines due to maintenance, failures or economic reasons. It is obviously unrealistic to require that AMGS should be trained for each possible combination of operating conditions and/or network topology because the dimensionality of the training pattern set would then become unacceptably large. The proposed solution to this problem is the parallel application of many individual neural nets, each focused on one, narrow, local subproblem and trained to handle a specific generator-shedding strategy.

Finally, it should be noted that the generator-shedding problem may also be approached using 'look-up' tables [15]. However, a table 'look-up' interpolation method, such as the nearest neighbour is only successful if the operating condition has previously been simulated or seen in practice. In the case where the current operating point is not included in the table the 'look-up' nearest neighbour method can introduce unacceptably high errors. This is due to the fact that 'look-up' table based mappings are constrained by a set of hyperplanes while in contrast, neural net mappings allow a high level of nonlinearity to exist in the decision hyperspace. Another advantage of neural nets is that their classification (recall) time is independent of the training set size and much shorter than for the nearest neighbour classifier [15].

4.1 Pattern generation strategy

The iterative process of calculating the amount of generation to be shed by the transient stability program is extremely time consuming. To facilitate these computations and collect the training data faster, we propose a general expression for the sensitivity of the energy margin to the amount of shed generation.

The energy margin has been developed [23] to indicate the degree of stability or instability and to rank the disturbances for different disturbance locations as

$$\Delta V = V^u - V^c \tag{14}$$

where V^u is the potential energy at the controlling UEP for the particular disturbance under investigation (critical energy); and $V^c = V^c_{pe} + V^c_{ke_{corrected}}$ is the transient energy function at the end of a disturbance (fault clearing time t^c) defined as a sum of the power system potential energy and kinetic energy corrected for the part that does not contribute to system separation [23].

In calculating the sensitivity of the energy margin with respect to the amount of shed generation we used two alternative approaches

(*a*) $V_{cr} = V^u$
(*b*) $V_{cr} = V_{pmax} = V_p(t^*)$

(based on the potential energy boundary surface-PEBS method [24]), where in our approach t^* is calculated iteratively as the time instant for which $dV_p/dt = 0$ as

$$\Delta t^{*(i)} = -[dV_p/dt]^{(i^*)}/[d^2V_p/dt^2]^{(i^*)}$$

where

$$\frac{dV_p}{dt} = \sum_{i=1}^{NG} \tilde{\omega}_i \left[E_i^2 G_{ii}^{pf} - P_{mi} + \sum_{\substack{j=1 \\ j\neq i}}^{NG} (C_{ij}^{pf} \sin\theta_{ij} + D_{ij}^{pf} \cos\theta_{ij}) \right]$$

$$\begin{aligned} \frac{d^2V_p}{dt^2} = {} & \sum_{i=1}^{NG} \tilde{\omega}_i \left[\sum_{\substack{j=1 \\ j\neq i}}^{NG} (C_{ij}^{pf} \cos\theta_{ij} - D_{ij}^{pf} \sin\theta_{ij}) \tilde{\omega}_{ij} \right] \\ & - \sum_{i=1}^{NG} \frac{1}{M_i} \left[\sum_{j=1}^{NG} (C_{ij}^{pf} \sin\theta_{ij} + D_{ij}^{pf} \cos\theta_{ij}) - P_{mi} \right]^2 \\ & + \frac{1}{M_T} \left[\sum_{i=1}^{NG} \sum_{j=1}^{NG} C_{ij}^{pf} \sin\theta_{ij} + D_{ij}^{pf} \cos\theta_{ij} - P_{mi} \right]^2 \end{aligned} \tag{15}$$

The sensitivity of the energy margin to shed generation is computed as

$$\xi = \frac{[V^{u(2)} - V^{c(2)}] - [V^{u(1)} - V^{c(1)}]}{\Delta P_m^{(2)} - \Delta P_m^{(1)}} \tag{16}$$

or

$$\xi^* = \frac{[V_{pmax}^{(2)} - V^{c(2)}] - [V_{pmax}^{(1)} - V^{c(1)}]}{\Delta P_m^{(2)} - \Delta P_m^{(1)}} \tag{17}$$

Here, $V^{u(1)}$, $V_{pmax}^{(1)}$, $V^{c(1)}$, $\Delta P_m^{(1)}$ are the critical energy (potential energy at the controlling unstable equilibrium point or potential energy at the PEBS crossing), the transient energy at the fault clearing time and the given amount of shed generation. The quantities $V^{u(2)}$, $V_{pmax}^{(2)}$, $V^{c(2)}$, $\Delta P_m^{(2)}$ are the same as those just described when the amount of shed generation is increased by a small value $\Delta P_m^{(2)} = \Delta P_m^{(1)} + \varepsilon \Delta P_m^{(1)}$.

Using the sensitivity of energy margin, the minimum amount of generation to be shed to prevent loss of synchronism can be calculated as the amount corresponding to the zero energy margin.

$$\Delta P_m^{min} = \Delta P_m^{(1)} - \frac{1}{\xi}[V_{cr}^{(1)} - V^{c(1)}] \tag{18}$$

The procedure for the appropriate calculation of the energy margin in the case of simultaneous line tripping and generator shedding at the fault clearing time is summarised in Fig. 8.

We calculated the minimum amount of generation to be shed for a three-phase fault at line 29–26 of the New England test system. The fault clearing policy is to perform simultaneously line tripping and generator shedding at the fault clearing time t^c, the results are shown in Table 1.

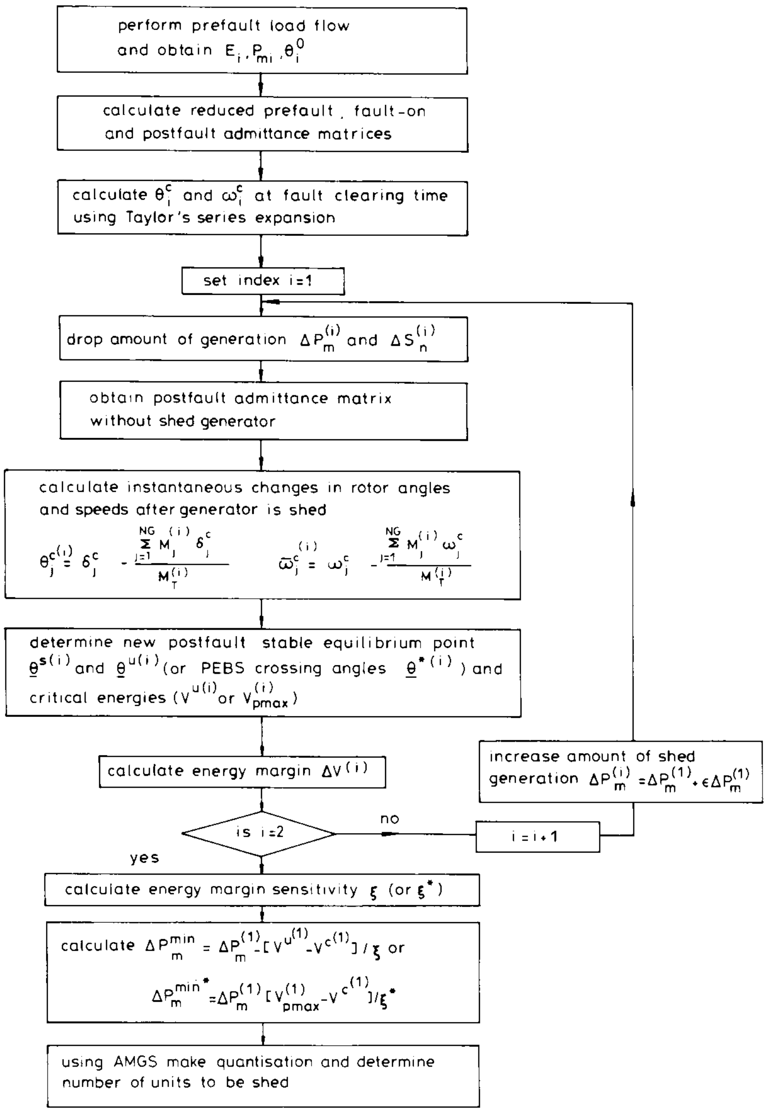


Fig. 8 *Use of energy margin sensitivity for fast data collection in AMGS*

Table 1: Results of calculation of required amount of generation to be shed in power station at bus 38 for a fault at line 29–26 with $t^c = 0.12$ using a sensitivity of energy margin

Load level	$V^{u(1)}$	$V^{(1)}_{pmax}$	ξ	ξ^*	ΔP_m^{min} Calculated by transient stability program	ΔP_m^{min} Calculated by ξ and $V^{u(1)}$	ΔP_m^{min*} Calculated by ξ^* and $V^{(1)}_{pmax}$
p.u.	p.u.	p.u.			MW	MW	MW
0.95	1.229	1.194	0.599	0.584	203	212	220
0.85	1.027	0.998	0.591	0.576	241	245	253
0.75	0.881	0.859	0.590	0.571	266	271	280
0.65	0.731	0.713	0.597	0.563	299	294	309
0.55	0.447	0.449	0.561	0.537	357	354	366

It is evident that the method which uses potential energy at the controlling unstable equilibrium point gives more accurate results, but results obtained using the PEBS method can be used for the purpose of training ANN as well, because the error is in an acceptable range.

4.2 Neural-net based generator shedding

The numerical experiments were carried out on the New England test system [25] which is shown in Fig. 9.

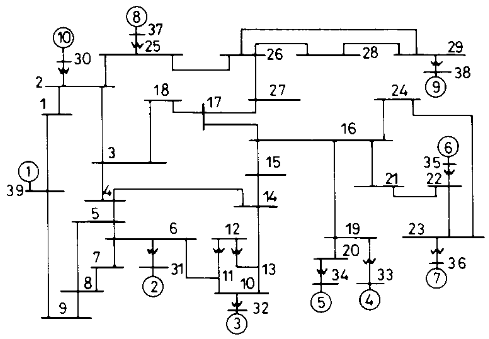


Fig. 9 *New England test system*

If a 3-phase short circuit fault occurs on line 29–26 and the faulted line is tripped in $t^c = 0.12$ s, we attempt to stabilise the system by shedding part of the generation at bus 38. We considered that the power station at bus 38 contained 8 × 125 MVA identical generator units and the generation shedding was executed simultaneously with the fault clearing. It should be noted, that a fault on line 29–26, with the fault clearing policy of isolating the fault by tripping the faulted line, causes a single mode of instability (generator 9 loses synchronism).

Using a transient stability program the minimum amounts of shed generation are determined, necessary to prevent loss of synchronism. To generate the training patterns we varied the loading level of the system in the range 0.3–1.0 relative to the nominal operating point in discrete steps (0.025 p.u. per step). The training set consisted of 22 10-dimensional patterns labelled with their corresponding control signals. The generator schedule used in obtaining the training set of patterns is given in the Appendix.

The performance of the neural net system for new cases, not presented during the training session i.e. for load levels 0.55 0.65 0.75 0.85 and 0.95 (p.u.) is illustrated in Table 2.

The joint activation FLN net used in this case with 10 input nodes and 45 enhancements was allowed to train itself until the least square error was reduced to 0.0000785 in 906 iterations with a learning rate of 0.7 and a momentum of 0.5. The maximum pattern error was 2.56×10^{-2} and the minimum pattern error 1.735×10^{-5}.

We observe that the estimated control signals agree with the results calculated by the transient stability program even when some of the lines were out of service (these cases were not presented to the neural net during the training phase).

Fig. 10 illustrates the effects of control actions when the system goes into instability without generator shedding. Fig. 11 shows that the system is stabilised when 3 × 125 MVA units are shed at bus 38 simultaneously with line tripping.

If a 3-phase short-circuit fault occurs on line 22–21 and the faulted line is tripped in $t^c = 0.27$ s, we attempt

Table 2: Results of calculation of required amount of generation to be shed in power station G9 for a fault at line 29–26 with $t^c = 0.12$ s

Load level	Lines out of service	Minimum amount of shed generation required to stabilise the system calculated by		Number of disconnected units needed to stabilise the system calculated by	
		Transient stability program	Neural-net system	Transient stability program	Neural-net system
p.u.		MW	MW	MVA	MVA
0.95	—	203	205	2 × 125	2 × 125
0.85	—	241	233	3 × 125	3 × 125
0.75	—	266	267	3 × 125	3 × 125
0.65	—	299	295	3 × 125	3 × 125
0.55	—	357	346	4 × 125	4 × 125
0.75	3–18	256	271	3 × 125	3 × 125
0.65	4–5	321	318	4 × 125	4 × 125
0.65	23–24	305	285	3 × 125	3 × 125
0.85	26–27	339	362	4 × 125	4 × 125
0.85	3–4	236	246	3 × 125	3 × 125
0.95	14–15	239	220	3 × 125	3 × 125
0.75	5–8 17–27	346	370	4 × 125	4 × 125

Table 3: Results of calculation of required amount of generation to be shed in power stations G6 and G7 simultaneously for fault at line 22–21 with $t^c = 0.27$ s

Load level	Minimum amount of shed generation required to stabilise system calculated by		Number of disconnected units needed to stabilise system calculated by	
	Transient stability program G6 + G7	Neural net system G6 + G7	Transient stability program	Neural net system
p.u.	MW	MW	MVA	MVA
0.95	290 + 245	304 + 257	2 × (4 × 125)	2 × (4 × 125)
0.85	182 + 147	178 + 143	2 × (3 × 125)	2 × (3 × 125)
0.75	2.4 + 1.8	5.8 + 4.4	2 × (1 × 125)	2 × (1 × 125)

to stabilise the system by shedding part of the generation at buses 35 and 36 simultaneously. It should be noted that the fault on line 22–21 causes a multiswing mode of instability (generators G1–G9 lose synchronism). The training set of data was generated as in the previous case, and results are shown in Table 3.

A FLN net used in this case was allowed to train until the least square error was reduced to 0.0000976 in 1155 iterations with a learning rate of 0.7 and a momentum of 0.5. The maximum pattern error was 3.11×10^{-2} and the minimum pattern error was 1.84×10^{-5}.

Fig. 12 shows the system going into instability without generator shedding while Fig. 13 shows that system is stabilised when 3 × 125 MVA units are simultaneously shed in power stations G6 and G7.

For comparison purposes we tested the proposed concept with FLN and with three architectures of the GDR neural net (single hidden layer with 20, 10 and 5 nodes, respectively). The personal computer IBM PC/AT was used, and results of comparison are shown in Fig. 14.

A wall-clock time needed for 100 iterations with FLN was 15 s while GDR neural nets needed 57, 23 and 16 s, respectively, for the same number of iterations. Enhanced input patterns of FLN, by means of functional transformations, allow a greatly simplified net architecture and higher learning rates than those of the generalised delta rule nets.

5 Conclusions

The associative memory system implemented on the computational platform of artificial neural networks is

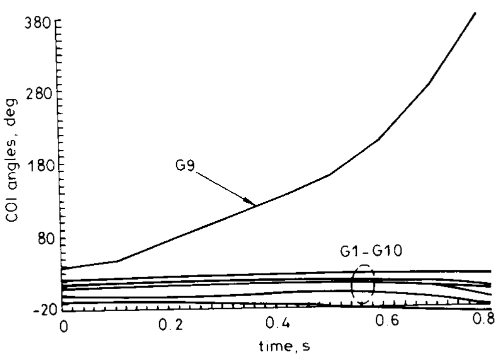


Fig. 10 *Simulation results for load level 0.75 p.u. a fault on line 29–26 with line tripping after 0.12 s and without generator shedding*

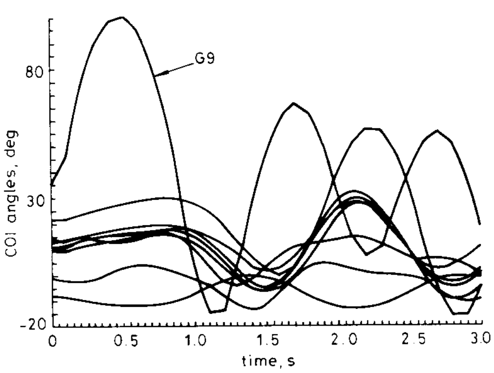


Fig. 11 *Simulation results for load level 0.75 p.u. fault on line 29–26 with line tripping after 0.12 s and with simultaneously shedding units 3 × 125 MVA in electric power station G9*

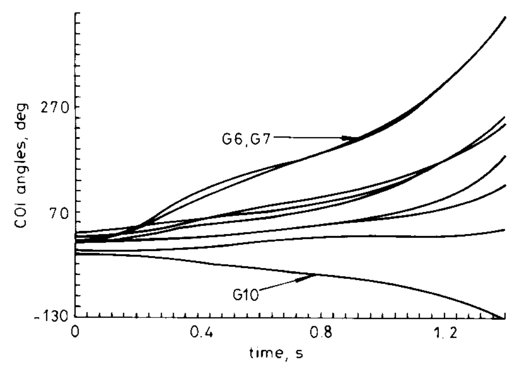


Fig. 12 *Simulation results for load level 0.85 p.u. fault on line 22–21 with line tripping after 0.27 s and without generator shedding*

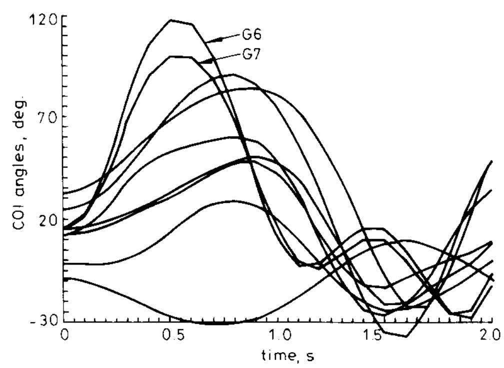


Fig. 13 *Simulation results for load level 0.85 p.u. fault on line 22–21 with shedding simultaneously 2 × (3 × 125) MVA in power stations G6 and G7*
$t^c = 0.27$ s

proposed for the fast determination of generator-shedding requirements in power systems. The critical amount of generator shedding required to prevent the loss of synchronism is determined using the FLN architecture within the supervised learning process. The complex mappings between the fault information and the amount of generation to be shed is efficiently learned by neural nets which are able to suggest the control action after appropriate training.

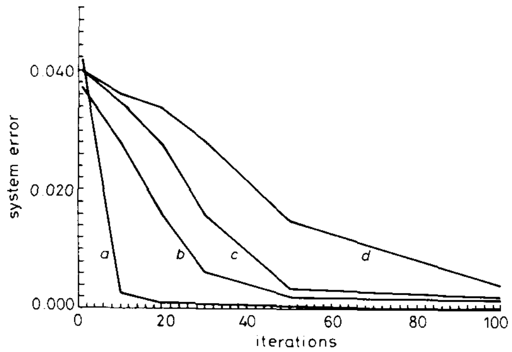


Fig. 14 *Comparison of rates of learning*
a FLN without hidden layers
b GDR-single hidden layer with 5 nodes
c GDR-single hidden layer with 10 nodes
d GDR-single hidden layer with 20 nodes

Data stored in a parallel hetero-associative memory can be always retrieved upon the presentation of a cue (input to the neural net system). A more interesting and useful capability of these systems is demonstrated when a new cue is presented, which does not match any of the previously stored data points. Owing to its ability to generalise, the memory will exhibit the outcome which will be equivalent to that obtained after running an entire power system simulation case.

The sensitivity of the transient energy margin with respect to shed generation is proposed as a fast method for the determination of the target control signal needed for ANN training. It has been shown that the PEBS method could be used as an alternative to controlling the unstable equilibrium concept to increase the computation, although the latter method provides more accurate results. The numerical examples presented demonstrate the generalisation capabilities of ANNs which allow calculations of generator-shedding requirements for different operating conditions.

6 References

1 IEEE task force on emergency control: 'Emergency control practices', *IEEE Trans.* 1985, **PAS-104**, (9), pp. 2336–2341
2 LUBKEMAN, D.L., and HEYDT, G.T.: 'The application of dynamic programming in a discrete supplementary control for transient stability enhancement of multimachine power systems', *IEEE Trans.* 1985, **PAS-104**, (9), pp. 2342–2348
3 FOUAD, A.A., GHAFURIAN, A., NODEHI, K., and MANSOUR, Y.: 'Calculation of generation-shedding requirements of the B.C. hydro system using transient energy function method', *IEEE Trans.* 1986, **PWRS-1**, (1), pp. 17–23
4 'Power system stability in Japan'. CRIEPI report, 1986
5 ANON.: 'Emergency prevention control system for bulk power systems', *Hitachi Rev.*, 1982, **31**, pp. 199–204
6 NISHIDA, S.: 'On-line identification of potential energy at the unstable equilibrium point and its application to adaptive emergency control of power system', *Elect. Engng. in Japan*, 1984, **104**, (4), pp. 113–123
7 OHURA, Y., MATSUZAWA, K., OHTSUKA, H., NAGAI, N., GOUDA, H., TAKEDA, S., and NISHIDA, S.: 'Development of a generator tripping system for transient stability augmentation based on the energy function method'. IEEE Winter Meeting, New York, 1986, p. 86, WM 117–6
8 RUMELHART, D.E., HINTON, G.E., and WILLIAMS, R.J.: 'Learning internal representation by error propagation in parallel distributed processing'. 'Explorations in the microstructures of cognition, vol. 1: foundations' (MIT Press, 1986), pp. 318–362
9 CYBENKO, G.: 'Approximation by superpositions of a sigmoidal function', *Math. Control Signals Syst.*, 1989, (2), pp. 303–314
10 HORNIK, K., STINCHCOMBE, M., and WHITE, H.: 'Multilayer feedforward networks are universal approximators', *Neural Networks*, 1989, (2), pp. 359–366
11 IRIE, B., and MIYAKE, S.: 'Capability of three-layered perceptrons'. Proceedings of IEEE Conference on neural networks, 1988, pp. 641–648
12 SOBAJIC, D.: 'Neural-nets for control of power systems'. PhD thesis, Case Western Reserve University, Cleveland, Ohio, 1988
13 PAO, Y.H.: 'Adaptive pattern recognition and neural networks' (Addison Wesley, 1989)
14 SOBAJIC, D.J., PAO, Y.H., and DOLCE, J.L.: 'On-line monitoring and diagnosis of power system operating conditions using artificial neural networks'. IEEE International Symposium of Circuits and Systems, Portland, OR, 1988
15 EL-SHARKAWI, M.A., and II, MARKS, R.J.: 'Artificial neural networks as operation aid for on-line static security assessment of power systems'. Power Systems Computation Conference, Graz, Austria, 1990
16 WEERASOORIYA, S., EL-SHARKAWI, M.A., DAMBORG, M. and II, MARKS, R.J.: 'Toward static security assessment of a large-scale power system using neural networks', *IEE Proc. C*, 1992, **139**, (1), pp. 64–70
17 SOBAJIC, D.J., and PAO, Y.H.: 'Artificial neural-net based security assessment for electric power systems', *IEEE Trans.* 1989, **PWRS-4**, (4), pp. 220–228
18 PAO, Y.H., and SOBAJIC, D.J.: 'Combined use of unsupervised and supervised learning for dynamic security assessment'. PICA'91, Baltimore, 1991
19 DILLON, T.S.: 'Artificial neural network applications to power systems and their relationship to symbolic methods', *Electrical Power & Energy Syst.* 1991, **13**, (2), pp. 66–72
20 MORI, H., TAMARU, Y., and TSUZUKI, S.: 'An artificial neural-net based technique for power system dynamics stability with Kohonen model'. PICA'91, Baltimore, 1991
21 DJUKANOVIC, M., SOBAJIC, D.J., and PAO, Y.H.: 'Neural-net based unstable machine identification using individual energy functions', *Electrical Power & Energy Syst.* 1991, **13**, (5), pp. 255–262
22 THOMAS, R.J., SAKK, E., HASHEMI, K., KU, B.J., and CHIANG. H.H.: 'On-line security screening using an artificial neural network'. IEEE-ISCAS-90, New Orleans, LA, 1990
23 VITTAL, V., RAJAGOPAL, S., FOUAD, A.A., EL-KADY, M.A., VACHEDI, E., and CARVALHO, V.F.: 'Transient stability analysis of stressed power systems using the energy function method', *IEEE Trans.* 1988, **PWRS-3**, (1), pp. 239–244
24 KAKIMOTO, N., OHNOGI, Y., MATSUDA, H., and SHIBUYA, W.: 'Transient stability analysis of large scale power systems by Lyapunov's direct method', *IEEE Trans.* 1984, **PAS-103**, (1), pp. 160–167
25 ATHAY, T., SHERKET, V.R., PODMORE, R., VIRMANI, S., and PUECH, C.: Conference on system engineering for power: emergency operating state control, Section IV, Davos, Switzerland, 1979. Also US Department of Energy Publication CONF-790904-PL, 1980

7 Appendix

The generation schedule used to obtain training pattern set

Load level	Active generation									
	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10
(p.u.)	MW									
1.0	250	563	650	632	508	650	560	540	830	1000
0.95	237	557	618	590	483	617	522	503	830	928
0.9	225	507	585	545	457	585	484	466	830	880
0.85	212	478	552	506	432	552	445	427	830	817
0.8	200	451	520	465	406	520	407	391	830	759
0.75	188	419	488	422	381	487	369	253	830	697
0.7	175	394	455	385	356	455	335	321	830	642
0.65	163	372	422	336	330	422	291	278	830	575
0.6	150	338	390	296	305	390	253	241	830	516
0.55	137	314	358	254	279	357	215	204	830	456
0.5	125	281	325	212	254	325	176	166	830	396